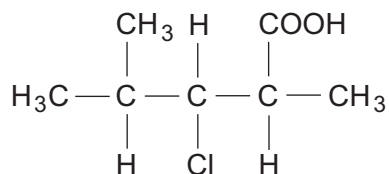


SECTION A*Answer all questions in the spaces provided.*

1. (a) Name the compound whose formula is shown below. [1]



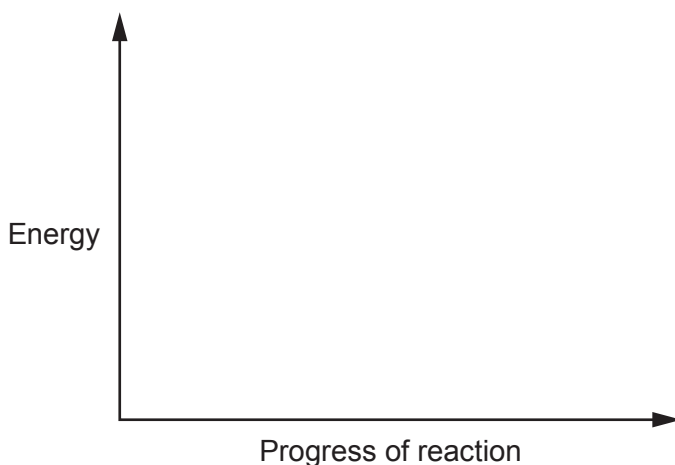
- (b) Draw the skeletal formula of the compound shown in (a). [1]

2. Respiration involves the release of energy by using foods. The equation for the respiration of glucose is shown below.

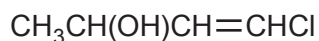


Photosynthesis is the process by which plants store energy in foods. It can be thought of as the reverse of respiration.

On the axes below sketch and label the energy profile for the **photosynthesis** process. You should label the enthalpy change, ΔH , and the activation energy, E_a . [2]



3. Draw a section of the addition polymer formed from the monomer below.



You should show **two** repeat units.

[1]

4. Bromine reacts with methanoic acid according to the equation below.



A student wanted to follow the rate of the reaction and mixed solutions of known concentration of bromine and methanoic acid.

Suggest **two** methods which the student could use to follow the rate of this reaction as it proceeds. [2]

Method 1

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Method 2

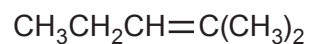
.....



5. (a) State what feature of a molecule gives rise to *E-Z* isomerism. [1]

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- (b) Does the molecule shown below show *E-Z* isomerism? Explain your answer. [1]



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6. On warming, copper(II) oxide reacts with aqueous methanoic acid. Write the equation for this reaction. [1]

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